Multi-source dataset

VitalDB
(Free Text)
"Gallbladder stone"

MOVER
(CMS-HCC term)
"Cholelithiasis"

INSPIRE
(ICD-10 code)
"K80.20"

13 clinical text fields

SNOMED CT Mapping

"Gallbladder stone"
SNOMED CT: 235919008

Matching Strategy

1 Exact match

Gallbladder stone → Gallbladder stone

2 Retrieval

Gallbladder stone →

3 LLM + RAG

+ → Selected concept

Semantic Clustering

1. Entity Extraction

→ cholelithiasis, pain, …

2. Synonym Normalization

gallstones, cholelithiasis, … → gallstone

3. Embedding & Clustering

Concept-Aligned Features

Concept-Clustered Features